

Notes: Murfin (JF, 2012)

The Supply-Side Determinants of Loan Contract Strictness

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Loan and borrower sample	4
3.2	Contract strictness	4
3.3	Lender defaults	4
3.4	Bank capitalization	4
3.5	Borrower outside options	5
4	Models and empirical specifications	5
4.1	Measure validation	5
4.2	Baseline borrower fixed effects specification	5
4.3	Distinct industry/geography specification	5
4.4	Capital and defaults	5
4.5	Legacy defaults and internal learning	6
4.6	Borrower outside options	6
5	Identification and empirical design	6
5.1	Core identification idea	6
5.2	Why default timing matters	6
5.3	Main identifying assumptions	6
5.4	Threats and responses	7
6	Tables and figures	8
6.1	Figure 1: Average contract strictness over time; Table I: Measure Validation	8
6.2	Table II: Summary Statistics and Sample Selection	8
6.3	Table III: Contract Strictness and Recent Defaults	10
6.4	Table IV: The Effects of Geographically and Industrially Distinct Defaults; Table V: Capital Effects and Recent Defaults	10
6.5	Table VI: Contract Strictness and Legacy Defaults; Table VII: Effects of Defaults on Internal and External Reputation	11
6.6	Table VIII: Contract Sensitivity and Lender Relationships; Table IX: Contract Sensitivity and Alternative Financing	12
7	Short summary	13
8	Conclusion	14

8.1	Core conclusion	14
8.2	Economic intuition	14
8.3	Incremental contribution revisited	14
8.4	Implications for credit cycles	14
8.5	Implications for borrowers	14
8.6	Limitations	14
8.7	Takeaway	14

1 Big picture

- **Question.** Holding borrower risk fixed, do lender-specific shocks affect the strictness of loan contracts?
- **Core object.** Contract strictness is measured as the ex ante probability that at least one financial covenant is violated, given the borrower's current accounting ratios and the covariance structure of covenant-ratio shocks.
- **Main result.** Banks write stricter contracts after recent payment defaults occur in their own loan portfolios.
- **Supply-side focus.** Prior covenant work emphasizes borrower risk. This paper shifts attention to lender learning, capital, and contracting tendencies.
- **Identification intuition.** If defaults to a lender's other borrowers affect contract terms for a new borrower - even when the defaulters are in different regions and industries - then the effect is unlikely to come only from new information about the current borrower.
- **Mechanism.** Defaults appear to make lenders update beliefs about their own screening ability or about loan-officer/committee processes, leading them to demand more contingent control.
- **Capital channel.** Weak bank capitalization also leads to tighter contracts, but lender defaults continue to matter after controlling for capital.
- **Heterogeneity.** The effect is stronger when borrowers have fewer bank relationships and weaker alternative financing options, and weaker when borrowers can switch banks or issue commercial paper.
- **Economic takeaway.** Contract design is not only a function of borrower risk; it also reflects lenders' recent experiences and internal beliefs about their screening technology.

2 Incremental contribution of the paper

1. **Creates a contract-level strictness measure.** Instead of counting covenants or studying one covenant at a time, Murfin constructs a multivariate probability of covenant violation that incorporates number of covenants, slack, scale, and covariance among accounting ratios.
2. **Moves covenant research to the supply side.** The paper asks how lender shocks influence contract design, whereas earlier work largely explains covenant structure with borrower risk and agency problems.
3. **Uses lender portfolio defaults as a shock to contracting preferences.** Recent defaults are lender-specific experiences that can change how a bank writes new contracts, even for unrelated borrowers.
4. **Separates learning from capital impairment.** The paper shows that recent defaults predict strictness even after controlling for bank capitalization and changes in bank capital, suggesting the result is not just balance-sheet damage.
5. **Tests internal versus external reputation.** By separating syndicated from bilateral loans and by examining legacy-loan defaults, the paper links tightening to internal screening/self-assessment rather than only external market reputation.
6. **Connects covenant strictness to borrower outside options.** The effects are concentrated among borrowers with fewer banking relationships and weaker nonbank financing access, showing that lender supply shocks matter most when borrowers cannot easily substitute away.
7. **Adds a new interpretation of covenant cycles.** Changes in loan terms over the credit cycle can arise from lenders' own experience and learning, not only from macro risk, borrower demand, or regulatory capital.

3 Data and measurement

3.1 Loan and borrower sample

The paper uses Loan Pricing Corporation’s DealScan database for syndicated and bilateral loans, merged to Compustat using the Chava-Roberts link file. The relevant observation is a loan package because covenants are reported at the package level. After restricting to contracts with covenant information and non-immediate violations, the covenant sample contains 2,642 loan packages. Because some packages have multiple lead arrangers, the lender-contract analysis uses borrower-lender observations.

Detailed interpretation: The unit of analysis matters. Using facilities instead of packages would duplicate the same covenant package across tranches and overstate precision.

3.2 Contract strictness

Strictness is designed to measure the probability that the lender obtains contingent control through a technical covenant violation. For a single ratio r with covenant threshold $\bar{r}$, a shock moves the ratio according to:

$$r' = r + \varepsilon.$$

For multiple covenants, let r be the vector of ratios and $\bar{r}$ the vector of thresholds. With shocks distributed as multivariate normal with covariance matrix Σ , strictness is:

$$STRICTNESS = 1 - F_N(r - \bar{r}; 0, \Sigma),$$

where F_N is the multivariate normal cumulative distribution function.

Detailed interpretation: The measure increases with more covenants, tighter slack, more volatile ratios, and lower covariance across covenant ratios. It is an economically interpretable index: an estimated probability of covenant violation.

3.3 Lender defaults

The main explanatory variable counts payment defaults on loans in the lead lender’s DealScan portfolio before the current loan contract is originated. The baseline measure focuses on defaults in the previous 90 days, and robustness uses alternative windows and legacy-loan definitions.

Detailed interpretation: These are lender-specific shocks. They are not technical covenant violations by the current borrower; they are payment defaults experienced by the lender in its loan book.

3.4 Bank capitalization

Bank capitalization is measured as bank shareholder equity divided by total assets. The paper studies both levels and changes in capitalization.

Detailed interpretation: If defaults matter only because they reduce bank capital, then controlling for bank capital and changes in capital should eliminate the default effect. The paper finds it does not.

3.5 Borrower outside options

The paper measures borrower outside options in two ways:

- prior bank relationships: the number of lenders the borrower used in up to four previous transactions;
- alternative financing: whether the borrower has commercial-paper access based on short-term ratings at or above A-2.

Detailed interpretation: The more outside options a borrower has, the easier it is to resist one lender’s tightened contract or switch to another funding source.

4 Models and empirical specifications

4.1 Measure validation

The validation regression tests whether the strictness measure predicts actual covenant violations:

$$Pr(Violation_i = 1) = \Phi(\alpha + \beta Strictness_i + \Gamma X_i + \epsilon_i).$$

A positive β means the ex ante probability measure captures real contract tightness.

Detailed interpretation: Table I shows strictness predicts violations better than simpler alternatives such as covenant count or slack in a single net-worth covenant.

4.2 Baseline borrower fixed effects specification

The main specification is:

$$Strictness_{i\ell t} = \alpha_i + \beta Defaults_{\ell,t-90:t} + \Gamma X_{i\ell t} + \delta_t + \lambda_{type} + \epsilon_{i\ell t},$$

where i is borrower, ℓ is lender, and t is contract date. The borrower fixed effect α_i compares different lenders’ contracts to the same borrower.

Detailed interpretation: Borrower fixed effects are central. They absorb time-invariant borrower risk and let the paper compare how lenders with different default experiences treat the same firm.

4.3 Distinct industry/geography specification

To rule out that lender defaults simply reveal risk about the current borrower, the paper excludes defaults by borrowers in the same state/country, the same one-digit SIC industry, or both:

$$Strictness_{i\ell t} = \alpha_i + \beta Defaults_{\ell,t}^{unrelated} + \Gamma X_{i\ell t} + FE + \epsilon_{i\ell t}.$$

Detailed interpretation: If a California high-tech borrower defaults and a West Virginia mining borrower gets a stricter contract from the same lender, the default is unlikely to be directly informative about the mining borrower.

4.4 Capital and defaults

The paper estimates variants of:

$$Strictness_{i\ell t} = \alpha_i + \beta Defaults_{\ell t} + \rho Capital_{\ell t} + \theta \Delta Capital_{\ell t} + \Gamma X_{i\ell t} + FE + \epsilon_{i\ell t}.$$

Detailed interpretation: A negative coefficient on changes in capital means lenders tighten contracts after capital falls. The default coefficient remains positive, so capital does not explain the full effect.

4.5 Legacy defaults and internal learning

The paper separates defaults according to how long ago the defaulted loans were originated:

$$Defaults_{\ell t}^{<1yr}, \quad Defaults_{\ell t}^{1-2yr}, \quad Defaults_{\ell t}^{2-5yr}, \quad Defaults_{\ell t}^{>5yr}.$$

Detailed interpretation: Defaults on recently originated loans are more informative about current screening quality than defaults on legacy loans. The stronger effect of recent-loan defaults supports the self-assessment or screening-learning channel.

4.6 Borrower outside options

The paper interacts recent defaults with borrower relationship breadth and alternative financing access. If lender supply shocks matter more when borrowers have fewer alternatives, the coefficient should be stronger in thin-relationship or non-CP-issuer samples.

Detailed interpretation: Tables VIII and IX show exactly this pattern: contract tightening is concentrated among borrowers with fewer bank relationships and weaker nonbank financing options.

5 Identification and empirical design

5.1 Core identification idea

The paper uses within-borrower variation in which lenders have recently experienced defaults. The identifying comparison is:

same borrower, same period risk controls, different lender default histories.

Detailed interpretation: This design is powerful because borrower fixed effects absorb persistent borrower risk. The remaining variation comes from lender-specific recent loan portfolio experiences.

5.2 Why default timing matters

Defaults in the prior 90 days are used because they are recent enough to affect beliefs and internal processes, while occurring before the contract is originated. The paper also tests other lag windows to show that the effect is concentrated in recent experience.

Detailed interpretation: Timing supports the supply-side interpretation. A lender cannot tighten a contract today in response to a default that has not yet occurred.

5.3 Main identifying assumptions

- Lender defaults are not perfectly correlated with unobserved changes in the risk of the current borrower after controls.
- Borrower fixed effects, ratings, covenant controls, loan year dummies, and loan-type dummies absorb the major demand-side risk differences.

- Defaults in unrelated industries/geographies are plausibly not direct news about the borrower.
- Capital controls absorb the mechanical balance-sheet channel from default losses.

5.4 Threats and responses

- **Unobserved borrower risk.** Addressed with borrower fixed effects and distinct industry/geography default tests.
- **Bank capital shocks.** Addressed by including capital levels and changes.
- **Common macro risk.** Addressed with time controls and aggregate default variables.
- **External reputation.** Tested through bilateral versus syndicated loans and legacy-loan defaults.
- **Borrower bargaining power.** Tested with relationship breadth and CP access heterogeneity.

Detailed interpretation: The design is not a randomized experiment, but the layered tests make the supply-side interpretation credible.

6 Tables and figures

6.1 Figure 1: Average contract strictness over time; Table I: Measure Validation

Read:

- Figure 1 plots average strictness against the Federal Reserve survey measure of tightening credit standards.
- The two series move together, with a correlation of about 0.6, validating strictness as a supply-side tightness measure.
- Table I shows that strictness predicts actual covenant violations, while covenant counts and single-covenant slack measures perform worse.

Economic message: The figure validates the time-series behavior of the measure, while Table I validates the contract-level economic meaning of the measure.

Detailed interpretation: Strictness is not just a mechanical index. It behaves like a lender-tightness measure over time and predicts actual lender control events within loans.

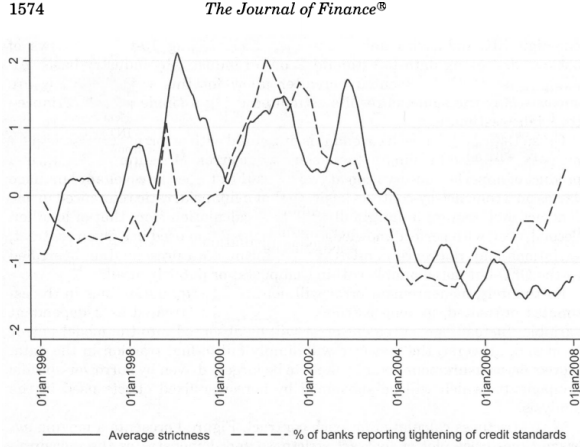


Figure 1. Average contract strictness over time, plotted against the percentage of respondents reporting tightening credit standards in the Federal Reserve's survey of senior loan officers. The moving average is calculated using a tent-shaped kernel over 180-day bandwidth, such that $STRICTNESS_t = \sum_{i=T-t \leq 180} [w_T(\sum_{i \in T} STRICTNESS_i)]$, where $w_T = \min[\frac{1 - |T-t|}{\sum_{i \in T} (1 - |T-i|)}, 0]$. Both plots are standardized.

(a) Figure 1: average contract strictness and Fed loan-officer tightening survey.

Figure 1 and Table I jointly validate the strictness measure.

Table I
Measure Validation

Probit regressions of borrower covenant violations occurring during the tenor of a given loan contract on three measures of loan strictness for that contract: strictness (the measure described in Section I, ranging from zero to one), the number of financial covenants, and the slack of the net worth or tangible net worth covenant (ATQ – LTQ – Covenant Level or ATQ – LTQ – INTANQ – Covenant Level, respectively, scaled by book assets). Covenant violation data come from Nini, Smith, and Sufi (2009). I consider only new covenant violations, consistent with the authors' instructions, by excluding violations in which the borrower had a violation within the past four quarters. Industry dummies are calculated at the one-digit SIC level. Ratings dummies are based on S&P long-term debt ratings (no rating is the base category). Covenant controls include the borrower's debt/tangible net worth, fixed charge coverage, current ratio, and ln(tangible net worth). Other variables are defined in the Appendix. Results are reported in terms of marginal effects evaluated at the mean of each variable. Standard errors are clustered by borrower, robust to heteroskedasticity, and are reported in parentheses. *** and ** signify results significant at the 1% and 5% levels, respectively.

Covenant Violations	I	II	III	IV	V	VI
Strictness	0.15*** (0.05)			0.22*** (0.07)	0.15*** (0.05)	0.20*** (0.08)
Number of financial covenants		0.01 (0.01)			0.00 (0.01)	0.02 (0.01)
Slack net worth covenant			-0.31** (0.14)	-0.11 (0.15)		-0.13 (0.15)
ln(maturity)	0.12*** (0.02)	0.12*** (0.02)	0.16*** (0.03)	0.15*** (0.03)	0.12*** (0.02)	0.15*** (0.03)
ln(amount)	-0.03** (0.01)	-0.03** (0.01)	-0.02 (0.02)	-0.02 (0.02)	-0.03** (0.01)	-0.02 (0.02)
Secured	0.05*** (0.02)	0.06*** (0.02)	0.07** (0.03)	0.06** (0.03)	0.05*** (0.02)	0.06** (0.03)
ln(number of participants)	0.00 (0.01)	0.01 (0.01)	0.00 (0.02)	0.00 (0.02)	0.00 (0.01)	-0.00 (0.02)
Borrower Z-score	-0.01** (0.00)	-0.01*** (0.00)	-0.01** (0.01)	-0.01** (0.01)	-0.01** (0.00)	-0.01** (0.01)
Observations	2,548	2,552	1,283	1,280	2,548	1,280
Log likelihood	-1,203	-1,211	-640	-634	-1,203	-633
Covenant controls	Yes	Yes	Yes	Yes	Yes	Yes
Ratings dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes

(b) Table I: probit validation of strictness using covenant violations.

6.2 Table II: Summary Statistics and Sample Selection

Read:

- Table II compares the full DealScan-Compustat sample with the covenant sample.

- The covenant sample contains 2,642 loan packages with mean strictness of about 22.5 percent and median strictness of about 17.5 percent.
- Lenders in the sample are large and typically well-capitalized, with mean capital around 7.5 percent.

Economic message: The table shows that the covenant sample is smaller and more information-rich than the full loan sample. It also documents the baseline distribution of firm, loan, and bank characteristics.

Detailed interpretation: Because strictness requires covenant data, sample selection matters. The table shows the sample differences and establishes the scale of the key variables.

Table II
Summary Statistics and Sample Selection

Summary statistics at the loan level for the merged DealScan–Compustat sample and the sub-sample for which covenants used to calculate loan contract strictness are reported. For loans with multiple lead arrangers, bank summary statistics represent the average of the lead arrangers.

	DealScan–Compustat Sample					
	<i>N</i>	Mean	<i>SD</i>	10 th	50 th	90 th
Firm characteristics						
Total assets (\$M)	22,020	3,515.27	11,993.31	51.71	599.86	8,268.60
EBITDA/assets	18,305	0.12	0.13	0.03	0.12	0.23
Market value of equity/Book liabilities	20,091	3.07	10.77	0.28	1.28	6.04
Has S&P long-term debt rating	22,789	0.43	0.50	0.00	0.00	1.00
S&P long-term debt rating	9,813	12.87	3.67	8.00	13.00	17.00
Altman <i>Z</i> -score	15,055	3.62	4.67	0.97	2.83	6.55
Loan characteristics						
Maturity (months)	20,942	49.06	289.52	12.00	42.00	82.00
Amount (\$M)	22,775	349.08	1,016.78	10.00	120.00	800.00
Secured	22,789	0.49	0.50	0.00	0.00	1.00
Number of participants	22,789	6.65	9.02	1.00	3.00	16.00
Number of lead arrangers	22,789	1.18	0.52	1.00	1.00	2.00
	DealScan–Compustat Covenant Sample					
	<i>N</i>	Mean	<i>SD</i>	10 th	50 th	90 th
Firm characteristics						
Total assets (\$M)	2,642	3,103.93	8,158.32	105.96	818.30	6,404.80
EBITDA/assets	2,642	0.15	0.07	0.07	0.14	0.24
Market value of equity/book liabilities	2,583	1.20	1.10	0.36	0.89	2.29
Has S&P long-term debt rating	2,642	0.49	0.50	0.00	0.00	1.00
S&P long-term debt rating	1,285	12.34	2.73	9.00	12.00	16.00
Altman <i>Z</i> -score	2,474	3.92	2.88	1.66	3.38	6.49
Loan characteristics						
Contract strictness	2,642	22.51	20.30	0.002	17.47	52.01
Maturity (months)	2,623	47.64	20.53	12.00	57.00	62.00
Amount (\$M)	2,642	411.35	773.46	27.50	200.00	950.00
Secured	2,642	0.51	0.50	0.00	1.00	1.00
Number of participants	2,642	9.25	8.83	1.00	7.00	20.00
Number of lead arrangers	2,642	1.23	0.43	1.00	1.00	2.00
Bank characteristics						
Lender total assets (\$BN)	2,504	589.79	475.89	83.58	450.56	1,337.91
Lender capitalization	2,481	7.51%	1.80%	5.29%	7.77%	9.31%
Defaults on lender portfolio—past 90 days	2,642	1.51	2.42	0.00	0.00	5.00

(continued)

The Supply-Side Determinants of Loan Contract Strictness 1579

Table II—Continued					
	DealScan–Compustat Covenant Sample				
	<i>N</i>	Mean	<i>SD</i>	10 th	50 th 90 th
% Loans arranged by Top 3 banks		37.52%			
% Loans arranged by Top 5 banks		48.52%			

(a) Table II, part 1: full DealScan–Compustat and covenant sample summaries.

(b) Table II, part 2: continued sample-selection information.

Table II summarizes the firm, loan, contract, and bank variables used in the analysis.

6.3 Table III: Contract Strictness and Recent Defaults

Read:

- Panel A shows that lender defaults in the prior 90 days are positively associated with contract strictness.
- A one-default increase in the lender's recent portfolio default count raises strictness by about 0.36 to 0.39 points in the baseline 90-day specifications.
- Panel B adds macro controls such as aggregate defaults, credit spreads, stock returns, and GDP growth; the lender-specific default effect remains.

Economic message: This is the paper's first main supply-side result: recent lender portfolio pain leads to stricter new contracts, even controlling for borrower fixed effects.

Detailed interpretation: The borrower fixed-effects setup means the same borrower can receive a stricter contract from a lender that recently experienced defaults. This is hard to explain with borrower risk alone.

Table III
Contract Strictness and Recent Defaults

Panels A and B present borrower fixed effects regressions of loan strictness as described in Section I (ranging from zero to 100), on the number of defaults in the 90 days prior to contracting and controls. Defaults on the lender's portfolio are calculated as the number of outstanding DealScan loan packages in which the lead arranger participated and for which the borrower's rating was changed to "Default" by the S&P ratings database during the period of interest. Covenant controls include the borrower's debt to tangible net worth, fixed charge coverage, current ratio, and ln(tangible net worth). Other variables are defined in the Appendix. Standard errors are clustered by borrower and lender, are robust to heteroskedasticity, and are reported in parentheses. ***, **, and * signify results significant at the 1%, 5%, and 10% levels, respectively.

Loan strictness	Panel A				
	I	II	III	IV	V
Defaults on lender portfolio—past 360 days	0.12** (0.05)				
Defaults on lender portfolio—past 90 days		0.38** (0.18)	0.38** (0.18)	0.36** (0.18)	0.39** (0.17)
Defaults on lender portfolio—90–180 days		0.13 (0.14)	0.14 (0.15)	0.12 (0.14)	
Defaults on lender portfolio—180–270 days		−0.09 (0.15)	−0.08 (0.15)		
Defaults on lender portfolio—270–360 days		0.07 (0.15)			
ln(maturity)	−0.88 (0.92)	−0.88 (0.93)	−0.89 (0.92)	−0.89 (0.92)	−0.89 (0.92)
ln(amount)	2.32** (1.05)	2.38** (1.04)	2.39** (1.04)	2.37** (1.04)	2.35** (1.05)
Secured	0.82 (1.87)	0.89 (1.87)	0.89 (1.87)	0.87 (1.87)	0.85 (1.86)
ln(number of participants)	1.28 (0.86)	1.30 (0.87)	1.27 (0.87)	1.28 (0.86)	1.29 (0.86)
Borrower Z-score	−1.39*** (0.48)	−1.39*** (0.48)	−1.39*** (0.48)	−1.39*** (0.48)	−1.40*** (0.48)
Observations	2,289	2,289	2,289	2,289	2,289
R ² (partial, excluding unreported fixed effects)	0.160	0.162	0.162	0.161	0.161
Ratings dummies	Yes	Yes	Yes	Yes	Yes
Borrower fixed effects	Yes	Yes	Yes	Yes	Yes
Covenant controls	Yes	Yes	Yes	Yes	Yes
Loan year dummies	Yes	Yes	Yes	Yes	Yes
Loan type dummies	Yes	Yes	Yes	Yes	Yes

(continued)

Table III—Continued

Loan strictness	Panel B			
	I	II	III	IV
Defaults on lender portfolio—past 90 days	0.34** (0.17)	0.32* (0.17)	0.33** (0.17)	0.35** (0.16)
ln(maturity)	−0.87 (0.98)	−0.92 (0.97)	−0.86 (0.98)	−0.89 (0.97)
ln(amount)	1.71 (1.07)	1.74* (1.05)	1.71 (1.05)	1.75* (1.05)
Secured	1.34 (1.87)	1.39 (1.85)	1.35 (1.89)	1.36 (1.86)
ln(number of participants)	1.52* (0.87)	1.53* (0.87)	1.51* (0.87)	1.48* (0.87)
Borrower Z-score	−1.45*** (0.48)	−1.44*** (0.48)	−1.45*** (0.48)	−1.45*** (0.48)
Aggregate defaults—past 90 days	0.29*** (0.06)	0.31*** (0.07)	0.29*** (0.06)	0.29*** (0.06)
Baa–Aaa credit spreads		−1.40 (2.49)		
S&P 500 return—past 90 days			1.03 (7.15)	
Quarterly GDP growth				39.46 (59.19)
Observations	2,289	2,289	2,289	2,289
R ² (partial, excluding unreported fixed effects)	0.138	0.139	0.138	0.138
Ratings dummies	Yes	Yes	Yes	Yes
Borrower fixed effects	Yes	Yes	Yes	Yes
Covenant controls	Yes	Yes	Yes	Yes
Loan year dummies	No	No	No	No
Loan type dummies	Yes	Yes	Yes	Yes

- (a) Table III, Panel A: recent defaults and contract strictness. (b) Table III, Panel B: macro controls and aggregate default controls.

Table III establishes the main positive relation between lender defaults and contract strictness.

6.4 Table IV: The Effects of Geographically and Industrially Distinct Defaults; Table V: Capital Effects and Recent Defaults

Read:

- Table IV excludes defaults related to the current borrower by state/country, one-digit SIC industry, or both.

- Defaults in unrelated industries and regions still predict stricter contracts.
- Table V adds lender capitalization and changes in capitalization; recent defaults remain significant.
- A decline in lender capital is also associated with tighter contracts.

Economic message: Table IV rules out the idea that defaults merely reveal borrower-sector risk. Table V shows that lender balance-sheet weakness matters but does not fully explain the default effect.

Detailed interpretation: The results support a supply-side learning channel: lenders react to their own loan outcomes even when those outcomes are not directly informative about the new borrower.

Table IV
The Effects of Geographically and Industrially Distinct Defaults

Table IV presents borrower fixed effects regressions. Recent default counts in Columns I, II, and III exclude defaults in the same state (or country for non-U.S. borrowers) as the contracting borrower, the same one-digit SIC code, or both, respectively. Covenant controls include the borrower's debt to tangible net worth, fixed charge coverage, current ratio, and ln(tangible net worth). Other variables are as defined in the Appendix. Standard errors are clustered by borrower and by lender, are robust to heteroskedasticity, and are reported in parentheses. ***, **, and * signify results significant at the 1%, 5%, and 10% levels, respectively.

Loan strictness	Different State/Country I	Different Industry II	Different Industry & State/Country III
Defaults on lender portfolio—past 90 days	0.42** (0.18)	0.47*** (0.17)	0.49*** (0.18)
ln(maturity)	-0.95 (0.93)	-0.97 (0.93)	-0.98 (0.93)
ln(amount)	2.39** (1.06)	2.42** (1.04)	2.36** (1.05)
Secured	0.83 (1.85)	0.88 (1.89)	0.81 (1.88)
ln(number of participants)	1.21 (0.86)	1.19 (0.84)	1.23 (0.85)
Borrower Z-score	-1.40*** (0.48)	-1.40*** (0.48)	-1.41*** (0.48)
Observations	2,275	2,275	2,275
R ² (partial, excluding unreported fixed effects)	0.17	0.170	0.172
Ratings dummies	Yes	Yes	Yes
Borrower fixed effects	Yes	Yes	Yes
Covenant controls	Yes	Yes	Yes
Loan year dummies	Yes	Yes	Yes
Loan type dummies	Yes	Yes	Yes

Table V
Capital Effects and Recent Defaults

Borrower fixed effects regressions. The dependent variable is loan strictness, as described in Section I. Lender capitalization is the ratio of shareholder equity to total assets held by the lender in the quarter in which the loan is originated. Covenant controls include the borrower's debt to tangible net worth, fixed charge coverage, current ratio, and ln(tangible net worth). Other variables are as defined in the Appendix. Standard errors are clustered by borrower and lender, are robust to heteroskedasticity, and are reported in parentheses. ***, **, and * signify results significant at the 1%, 5%, and 10% levels, respectively.

Loan Strictness	I	II	III
Defaults on lender portfolio—past 90 days	0.44** (0.19)	0.43** (0.18)	0.41** (0.18)
Lender capitalization (<i>t</i>)	-0.40* (0.22)		
ΔLender capitalization (<i>t</i> + 1)			0.46 (0.40)
ΔLender capitalization (<i>t</i>)		-1.18*** (0.24)	-1.21*** (0.28)
ΔLender capitalization (<i>t</i> - 1)			-0.62 (0.41)
ln(maturity)	-1.00 (0.93)	-0.98 (0.96)	-1.10 (0.95)
ln(amount)	2.23** (1.13)	2.34** (1.14)	2.48** (1.18)
Secured	0.03 (1.93)	0.13 (1.93)	0.15 (1.98)
ln(number of participants)	1.28 (0.88)	1.26 (0.88)	1.10 (0.89)
Borrower Z-score	-1.81*** (0.65)	-1.82*** (0.65)	-1.84*** (0.65)
Observations	2,059	2,059	2,022
R ² (partial, excluding unreported fixed effects)	0.178	0.181	0.186
Ratings dummies	Yes	Yes	Yes
Borrower fixed effects	Yes	Yes	Yes
Covenant controls	Yes	Yes	Yes
Loan year dummies	Yes	Yes	Yes
Loan type dummies	Yes	Yes	Yes

Using specifications otherwise identical to Panel A of Table III (only now considering contracts for which there are data on current and lagged lender capitalization), I find that bank capital is strongly associated with contract strictness, again, conditional on borrower risk and economic conditions. The regression reported under Column I controls for the level of capitalization, whereas Column II considers the effects of changes to the lender's capitalization ratio. The level of capitalization is negative and significant at the 10% level.

(a) Table IV: geographically and industrially distinct defaults. (b) Table V: lender capital and default effects.

Tables IV and V separate borrower-risk, capital, and lender-experience channels.

6.5 Table VI: Contract Strictness and Legacy Defaults; Table VII: Effects of Defaults on Internal and External Reputation

Read:

- Table VI shows that defaults on loans originated in the last year have the strongest effect on new contract strictness.
- Defaults on legacy loans more than five years old have weaker effects.
- Table VII compares syndicated and bilateral loans to distinguish external reputation from internal screening channels.

- The bilateral interaction is not large or significant, suggesting the effect is not mainly driven by public syndicate-market reputation.

Economic message: Recent-loan defaults are most informative about current screening processes. The evidence favors internal learning about screening ability over pure external reputation.

Detailed interpretation: The timing of the defaulted loan's origination is a clever mechanism test. A recent default on a recently originated loan says more about the lender's current screening quality than a default on an old legacy loan.

Table VI
Contract Strictness and Legacy Defaults

Borrower fixed effects regressions. Defaults are sorted based on whether the defaulting loans were originated in the year prior to, between 1 and 2 years prior to, between 2 and 5 years prior to, or more than 5 years prior to the current contract. Column V tests the differential effects of defaults on recently originated loans (<1 year) and defaults on legacy loans (>5 years), each comprising roughly 10% of all defaults. Covenant controls include the borrower's debt to tangible net worth, fixed charge coverage, current ratio, and ln(tangible net worth). Other variables are as defined in the Appendix. Standard errors are clustered by borrower and lender, are robust to heteroskedasticity, and are reported in parentheses. ***, **, and * signify results significant at the 1%, 5%, and 10% levels, respectively.

Loan Strictness	I	II	III	IV	V
(i) Lender defaults (loans < 1-year old)	1.85*** (0.63)				1.85*** (0.63)
(ii) Lender defaults (1-year old < loans < 2-year old)		0.51 (0.41)			
(iii) Lender defaults (2-year old < loans < 5-year old)			0.49* (0.26)		
(iv) Lender defaults (loans > 5-year old)				0.19 (0.53)	0.21 (0.55)
(i) – (iv)					1.64*
ΔLender capitalization	–1.26*** (0.26)	–1.24*** (0.25)	–1.15*** (0.24)	–1.21*** (0.25)	–1.25*** (0.25)
ln(maturity)	–0.88 (0.95)	–0.96 (0.95)	–0.94 (0.96)	–0.95 (0.98)	–0.90 (0.97)
ln(amount)	2.20* (1.13)	2.37** (1.14)	2.30** (1.14)	2.33** (1.16)	2.21* (1.14)
Secured	0.08 (1.96)	0.04 (1.93)	0.15 (1.92)	0.09 (1.92)	0.09 (1.95)
ln(number of participants)	1.30 (0.86)	1.21 (0.87)	1.25 (0.88)	1.20 (0.87)	1.29 (0.86)
Borrower Z-score	–1.77*** (0.66)	–1.79*** (0.67)	–1.81*** (0.65)	–1.78*** (0.67)	–1.78*** (0.66)
Observations	2,059	2,059	2,059	2,059	2,059
R ² (partial, excluding unreported fixed effects)	0.182	0.179	0.180	0.178	0.182
Ratings dummies	Yes	Yes	Yes	Yes	Yes
Borrower fixed effects	Yes	Yes	Yes	Yes	Yes
Covenant controls	Yes	Yes	Yes	Yes	Yes
Loan year dummies	Yes	Yes	Yes	Yes	Yes

(a) Table VI: recent and legacy loan default effects.

Table VII
Effects of Defaults on Internal and External Reputation

Borrower fixed effects regressions. The interaction of interest is between defaults on the lender's portfolio in the 90 days leading up to contracting and the indicator for whether the current loan is syndicated or bilateral, where *Bilateral* is equal to one if DealScan reports the distribution method as "sole lender" or "bilateral." Covenant controls include the borrower's debt to tangible net worth, fixed charge coverage, current ratio, and ln(tangible net worth). Other variables are as defined in the Appendix. Standard errors are clustered by borrower and lender, are robust to heteroskedasticity, and are reported in parentheses. *** and ** signify results significant at the 1% and 5% levels, respectively.

Loan Strictness	
Defaults on lender portfolio—past 90 days	0.42** (0.18)
Defaults on lender portfolio—past 90 days × Bilateral	0.92 (1.06)
Bilateral	–4.49 (4.13)
ΔLender capitalization	–1.16*** (0.25)
ln(maturity)	–1.01 (0.95)
ln(amount)	2.30** (1.13)
Secured	0.20 (1.92)
ln(number of participants)	1.21 (0.87)
Borrower Z-score	–1.82*** (0.65)
Observations	2,059
R ² (partial, excluding unreported fixed effects)	0.183
Ratings dummies	Yes
Borrower fixed effects	Yes
Covenant controls	Yes
Loan year dummies	Yes
Loan type dummies	Yes

(b) Table VII: syndicated versus bilateral loans.

Tables VI and VII examine whether defaults affect contracts through internal learning or reputation.

6.6 Table VIII: Contract Sensitivity and Lender Relationships; Table IX: Contract Sensitivity and Alternative Financing

Read:

- Table VIII splits borrowers by the number of prior lender relationships.
- Defaults strongly affect strictness for borrowers with below-median bank relationships, but not for those with many lender relationships.
- Table IX splits borrowers by commercial paper access.
- Defaults raise strictness for non-CP issuers but not for CP issuers.

Economic message: Lender supply shocks bind most when borrowers have weak outside options. Borrowers with many lender relationships or CP access can avoid or resist harsher contract terms.

Detailed interpretation: These heterogeneity tests add bargaining power to the main mechanism. A lender may want stricter terms after defaults, but only some borrowers have to accept them.

Table VIII
Contract Sensitivity and Lender Relationships

Borrower fixed effects regressions. To estimate the breadth of lender relationships available to a borrower, I count the number of banks that have lent to a given borrower, going back up to four transactions. For borrowers with less than four prior deals, the number of lenders is scaled by the number of prior loans observed, up to four. Columns I and II split the sample into borrowers for which the number of lenders used in the prior four transactions was less than or greater than median. Covenant controls include the borrower's debt to tangible net worth, fixed charge coverage, current ratio, and $\ln(\text{tangible net worth})$. Other variables are as defined in the Appendix. Standard errors are clustered by borrower and lender, are robust to heteroskedasticity, and are reported in parentheses. ***, **, and * signify results significant at the 1%, 5%, and 10% levels, respectively.

Loan strictness	Number of Lender Relationships		I-II
	≤median I	>median II	
Defaults on lender portfolio—past 90 days	0.97*** (0.27)	0.08 (0.21)	0.89***
ΔLender capitalization	−1.20* (0.61)	−1.30*** (0.36)	
$\ln(\text{maturity})$	−1.74 (1.44)	0.27 (0.81)	
$\ln(\text{amount})$	2.00 (1.50)	−0.01 (1.42)	
Secured	1.98 (2.65)	0.63 (2.09)	
$\ln(\text{number of participants})$	1.91 (1.41)	0.79 (1.28)	
Borrower Z-score	−1.48** (0.66)	−4.62*** (0.84)	
Observations	926	864	
R^2 (partial, excluding unreported fixed effects)	0.233	0.320	
Ratings dummies	Yes	Yes	
Borrower fixed effects	Yes	Yes	
Covenant controls	Yes	Yes	
Loan year dummies	Yes	Yes	
Loan type dummies	Yes	Yes	

(a) Table VIII: contract sensitivity by lender relationship breadth.

Table IX
Contract Sensitivity and Alternative Financing

Borrower fixed effects regressions. Column II examines a subsample of borrowers with short-term ratings at or above A-2 as these firms have access to CP markets. Column I examines borrowers without short-term ratings and those rated weaker than A-2. Covenant controls include the borrower's debt to tangible net worth, fixed charge coverage, current ratio, and $\ln(\text{tangible net worth})$. Other variables are as defined in the Appendix. Standard errors are clustered by borrower and lender, are robust to heteroskedasticity, and are reported in parentheses. *** and ** signify results significant at the 1% and 5% levels, respectively.

Loan strictness	Non-CP Issuer I	CP Issuer II	I-II
Defaults on lender portfolio—past 90 days	0.50** (0.25)	−0.28 (0.22)	0.78**
ΔLender capitalization	−1.41*** (0.36)	−0.83*** (0.31)	
$\ln(\text{maturity})$	−0.50 (1.23)	0.00 (1.03)	
$\ln(\text{amount})$	2.80** (1.37)	3.14** (1.44)	
Secured	−0.25 (1.95)	1.96 (3.05)	
$\ln(\text{number of participants})$	1.50 (1.04)	−1.77 (1.58)	
Borrower Z-score	−1.87*** (0.71)	0.49 (0.83)	
Observations	1,674	366	
R^2 (partial, excluding unreported fixed effects)	0.175	0.526	
Ratings dummies	Yes	Yes	
Borrower fixed effects	Yes	Yes	
Covenant controls	Yes	Yes	
Loan year dummies	Yes	Yes	
Loan type dummies	Yes	Yes	

On the one hand, borrowers with access to the CP market benefit directly from the outside option provided by the CP issuance and therefore are less likely to accept nonmarket terms from their lenders. This idea follows several

(b) Table IX: contract sensitivity by commercial-paper access.

Tables VIII and IX show that borrower outside options moderate lender supply effects.

7 Short summary

- The paper develops a multivariate probability-based measure of loan contract strictness.
- Strictness predicts actual covenant violations, validating the measure.
- Contract strictness varies strongly over time and correlates with Fed survey measures of bank lending standards.
- Lenders write stricter contracts after recent defaults in their own loan portfolios.
- The effect survives borrower fixed effects, borrower risk controls, loan controls, time controls, and macro controls.
- It remains when defaults in the same geography or industry are excluded.
- Bank capitalization matters, but it does not fully explain recent-default effects.
- Defaults on recently originated loans matter more than defaults on old legacy loans.
- The effect is strongest for borrowers with weaker outside options, such as fewer bank relationships and no CP access.
- The main conclusion is that lenders' own experiences shape contract strictness, not just borrower characteristics.

8 Conclusion

8.1 Core conclusion

Murfin shows that loan contract strictness has a supply-side component. Lenders do not write the same contract for a given borrower regardless of their own experiences. After defaults occur in a lender's loan portfolio, the lender tightens new loan contracts, even when borrower risk is held fixed.

8.2 Economic intuition

Defaults may make banks question their screening ability, internal models, loan officers, or underwriting procedures. A stricter covenant package gives the bank more contingent control and more opportunities to renegotiate if the borrower deteriorates. When banks feel less confident about screening, stricter covenants are a way to shift monitoring and control rights earlier into the loan contract.

8.3 Incremental contribution revisited

The paper's contribution is twofold. First, it provides a contract-level strictness measure that uses the probability of covenant violation rather than simple covenant counts. Second, it shows that lender-side shocks affect covenant design. Together, these contributions move the covenant literature from borrower-side agency problems toward lender experience, learning, and bargaining power.

8.4 Implications for credit cycles

The results suggest that covenant cycles can reflect lender learning and lender balance sheets. During periods after defaults, banks may collectively tighten covenants not only because borrower risk rises but also because their own recent experiences make them more cautious. This offers a microfoundation for the co-movement between loan strictness and credit conditions.

8.5 Implications for borrowers

Borrowers with many banking relationships or access to commercial paper can reduce exposure to one lender's tightening. Borrowers with few relationships are more vulnerable to lender-specific shocks. This means that financial flexibility and relationship breadth can affect not only loan availability and pricing, but also the strictness of control rights embedded in debt contracts.

8.6 Limitations

The paper is not a randomized experiment. Lender defaults may still be correlated with unobserved lender-specific or borrower-specific conditions. The strictness measure is estimated from stylized assumptions about accounting-ratio shocks. Covenant data are also available for only a subset of loans. However, the layered evidence - borrower fixed effects, unrelated-default tests, capital controls, legacy-default tests, and outside-option heterogeneity - makes the supply-side interpretation persuasive.

8.7 Takeaway

The key takeaway is that debt contracts are shaped by lender beliefs and lender experiences. A loan covenant package is not only a response to borrower risk; it is also a reflection of the lender's recent history and confidence in its own screening technology.